

Exercise 2.2

sums



Question 1 – Parts (i) & (ii)



Evaluate the following if $z = 5 - 2i$ and $w = -1 + 3i$

(i) $z + w$

$$\begin{aligned}(5 - 2i) + (-1 + 3i) \\&= (5 - 1) + (-2 + 3)i \\&= 4 + i\end{aligned}$$

Answer: $4 + i$

(ii) $z - iw$

$$\begin{aligned}\text{First: } iw &= i(-1 + 3i) = -i + 3i^2 \\&= -i - 3 = -3 - i \\ \text{Then: } z - iw &= (5 - 2i) - (-3 - i) \\&= 8 - i\end{aligned}$$

Answer: $8 - i$

Question 1 – Parts (iii) & (iv)



Evaluate the following if $z = 5 - 2i$ and $w = -1 + 3i$

(iii) $2z + 3w$

$$2z = 2(5 - 2i) = 10 - 4i$$

$$3w = 3(-1 + 3i) = -3 + 9i$$

$$2z + 3w = 7 + 5i$$

Answer: $7 + 5i$

(iv) zw

$$(5 - 2i)(-1 + 3i)$$

$$= -5 + 15i + 2i - 6i^2$$

$$= -5 + 17i + 6$$

$$= 1 + 17i$$

Answer: $1 + 17i$

Question 1 – Parts (v) & (vi)



Evaluate the following if $z = 5 - 2i$ and $w = -1 + 3i$

(v) $z^2 + 2zw + w^2$

Since $z^2 + 2zw + w^2 = (z + w)^2$

From (i): $z + w = 4 + i$

$(4 + i)^2 = 16 + 8i + i^2 = 15 + 8i$

Answer: $15 + 8i$

(vi) $(z + w)^2$

$z + w = 4 + i$

$(4 + i)^2 = 15 + 8i$

Answer: $15 + 8i$

Question 2 – Argand Diagram



Given $z = 2 + 3i$, represent the following points

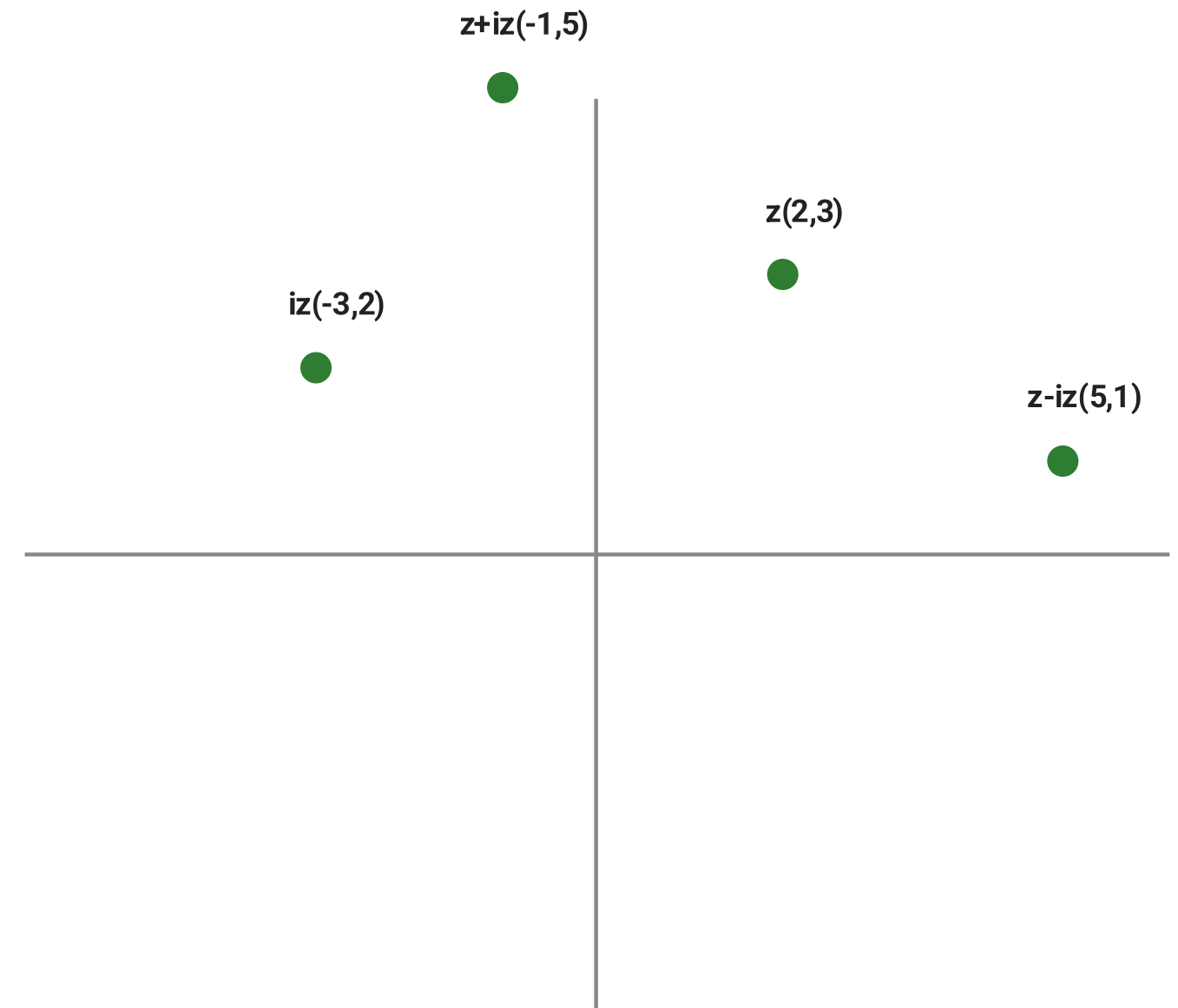
(i) $z, iz, z + iz$

- $z = 2 + 3i \rightarrow$ Point $(2, 3)$
- $iz = i(2 + 3i) = -3 + 2i \rightarrow$ Point $(-3, 2)$
- $z + iz = -1 + 5i \rightarrow$ Point $(-1, 5)$

(ii) $z, z - iz$

- $z = 2 + 3i \rightarrow$ Point $(2, 3)$
- $z - iz = 5 + i \rightarrow$ Point $(5, 1)$

Argand Diagram



Question 3 — Find the values of x and y



Given $(3-i)x - (2-i)y + 2i + 5$ and $2x + (-1+2i)y + 3 + 2i$ are equal

Expand the left side:

$$\begin{aligned} &(3x - ix) - (2y - iy) + 5 + 2i \\ &= (3x - 2y + 5) + (-x + y + 2)i \end{aligned}$$

Right side:

$$(2x - y + 3) + (2y + 2)i$$

Equating real & imaginary parts:

$$\text{Real: } 3x - 2y + 5 = 2x - y + 3 \Rightarrow x - y = -2$$

$$\text{Imag: } -x + y + 2 = 2y + 2 \Rightarrow x + y = 0$$

Solving the system:

$$x - y = -2$$

$$x + y = 0$$

$$\text{Adding: } 2x = -2 \Rightarrow x = -1$$

$$\text{From } x + y = 0 \Rightarrow y = 1$$

Answer: $x = -1, y = 1$